

Hierarchical Closed-Loop Cooperative Localization (HCCC): A Resource-Constrained, Cross-Layer Design for Best-Achievable-Accuracy Indoor Positioning on Commodity Smartphones

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Abstract

Cooperative localization has traditionally been framed as an accuracy-maximization problem. We argue—and investigate—the alternative design objective of achieving the *best achievable* localization accuracy under constrained computation, communication, and sensing resources. We propose *Hierarchical Closed-Loop Cooperative Localization* (HCCC), an initial, fully decentralized, cross-layer framework that instantiates this resource-constrained principle through five coupled modules. A reproducible, dependency-free simulator indicates meter-level accuracy (median 2.9–3.5 m) at near-linear complexity; the PDR front-end and the complete cooperative protocol are further validated on public real-phone IMU traces (RIDI), where a single trusted correction reduces mean trajectory error from 12.5 m to 8.1 m (−35%), and the full pipeline runs end-to-end across 16 real phones with a modest +2.2% gain bounded by real PDR drift. This is preliminary work: we outline the design principle, an initial architecture, and early empirical validation, and identify the open problems—notably Sybil-resistant admission and learned PDR priors—that define the research agenda.

Keywords

Indoor localization, Cooperative localization, Pedestrian dead reckoning, Resource-constrained design, Smartphone, Edge AI

1. Introduction

Cooperative localization has, until now, been framed chiefly as an accuracy-optimization problem: given a set of nodes, estimate each position as precisely as possible by fusing every available measurement. Yet as the node population scales to the hundreds or thousands of commodity phones that fill a building, the limiting factor is rarely the localization algorithm itself. It is the *resource cost* of cooperation—the computation of a growing joint state, the communication and energy of peer ranging and exchange, and the coordination overhead of keeping every node in the loop. Flat cooperative schemes that maximize accuracy pay this cost super-linearly ($\mathcal{O}(N^3)$ joint-state updates [1]), which is precisely why they never reach commodity phones.

We therefore argue for a shift in the objective of cooperative localization: from *maximizing accuracy* to achieving the *best achievable accuracy under constrained resources*. The right question is no longer “how accurate can cooperation make us?” but “what is the highest accuracy attainable at a bounded computation, communication, and sensing budget, and how do we get there?” Under this principle, the goal is not that *every* node cooperates with *every* other node; it is to extract the largest accuracy gain from the smallest cooperative cost. This paper adopts that principle as its central thesis:

Cooperative localization should not pursue universal cooperation; it should pursue the best achievable localization accuracy at the lowest computation, communication, and sensing cost under limited resources.

IPIN-WCAL 2026: Workshop for Computing & Advanced Localization at the Sixteenth International Conference on Indoor Positioning and Indoor Navigation, October 5–8, 2026, Rome, Italy

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Notably, the cooperative-localization literature has almost exclusively treated accuracy as the quantity to maximize—through richer graphs, better filters, or stronger physical-layer signals [1, 2, 3]—while the *explicit* joint optimization of accuracy against a fixed computation, communication, and energy budget has received little attention. HCCC’s contribution is therefore less a new estimator and more a reformulation of what the field should be optimizing: the accuracy–cost trade-off itself becomes the design variable.

HCCC is the cross-layer architecture we propose to instantiate this principle. It couples five modules—perception, topology, consensus, hierarchy, and routing—each adapted from prior art but coupled so that the whole stays linear in cost while retaining a cooperative accuracy gain: its topology pruning and hierarchical averaging keep peer selection at $\mathcal{O}(V + E)$ and interference at $\mathcal{O}(\sqrt{N})$, turning the accuracy–cost trade-off into a controllable knob.

Positioning of this work. This is a *work-in-progress* report. We present (i) a resource-constrained design principle we believe is under-explored, (ii) an *initial* architectural framework (HCCC) that realizes it, and (iii) *early* empirical validation on both simulation and public real-phone traces. We make no claim of a deployed physical prototype, nor of outperforming specialized infrastructure such as UWB; the open problems that remain are stated explicitly in section 7.

2. Related Work

PDR regresses step length from IMU [4] or, learned, velocity from raw IMU (RoNIN [5]). Cooperative localization [1] and centralized smartphone particle filters [6] show cooperation helps but inherit flat-network cost. Crowd-sourced, infrastructure-light methods such as PeepLoc [7] bootstrap Wi-Fi anchors through PDR, while recent Wi-Fi RTT [2, 8] and 5G NR [3] approaches add strong new signals but remain anchor- or base-station-dependent with super-linear joint-estimation cost. HCCC’s emphasis is the complexity- and energy-controlled *cooperation layer* that makes cross-layer fusion practical on resource-constrained phones regardless of the ranging signal.

3. System Model and Resource-Constrained Objective

Each phone v maintains a PDR prior $\hat{\mathbf{p}}_v$ with error σ_0 , exchanges ranging with peers within a density-aware radius, and participates in three coupled layers: **Topology**—the sensing radius shrinks in crowds to bound peer count; **Consensus**—asynchronous *Cycles* give loop-closure corrections; **Hierarchy**—clusters take the average position, cancelling hardware noise by the Law of Large Numbers; **Routing**—a depth-first *candle* walk projects a confidence-decaying trust chain $W(h) = \gamma^h$ from GPS anchors into deep indoor space, with pruning so each node is visited once. HCCC is fully decentralized and permissionless.

What “best achievable” means. The phrase is only meaningful relative to a fixed resource envelope. We therefore make the budget explicit. Let a node’s per-round *communication* budget be B_{comm} scalars, its per-session *energy* budget be E_{max} , and its per-update *CPU* budget bound the update latency. The design objective is

$$\min_{\{\mathbf{x}_i\}, \mathcal{G}} \mathbb{E}[\|\mathbf{x}_i - \mathbf{x}_i^*\|^2] \quad \text{s.t.} \quad \text{cost}(\mathcal{G}) \leq B_{\text{comm}}, \quad E_i \leq E_{\text{max}}, \quad \forall i, \quad (1)$$

rather than unconditional accuracy maximization. Concretely, HCCC enforces this envelope by (a) reporting a group centroid plus variance—3 scalars—instead of $2N$ raw coordinates (so B_{comm} stays constant as N grows), (b) batching updates into a $T = 2$ s window that cuts CPU wake-ups by $\approx 10\times$ versus continuous updating, and (c) contracting the sensing radius in crowds to cap the active peer count at 3–6. “Best achievable accuracy” is thus the accuracy the cooperative layer can reach *without exceeding* these envelopes.

Table 1

HCCC modules vs. prior art: adapted vs. contributed.

Module	Prior art	Relation	New in HCCC
Perception	RoNIN	Adapt	coupling
Topology	SPAWN	Adapt	radius→ $\mathcal{O}(V + E)$
Consensus	Olfati-Saber	Adapt	batching
Hierarchy	Seco & Jiménez	Adapt	LLN comp.
Routing	Wymeersch	Extend	prune→ $\mathcal{O}(V + E)$

Definition 1 (Best-achievable accuracy). The *best-achievable accuracy* is the minimum localization error attainable under fixed computation, communication, and sensing budgets.

4. The HCCC Framework

HCCC is organized as five layers traversed by a single data-flow pipeline: sensor input → perception → topology → consensus → hierarchical aggregation → backbone routing.

- **Perception.** A lightweight on-device model classifies carrying mode and performs online step-length calibration; the PDR prior is personalized before being shared.
- **Topology.** A density-aware radius R_i bounds the peer graph *before* routing prunes it, preserving the $\mathcal{O}(V + E)$ peer-selection bound.
- **Consensus.** Local neighbors form a closed *Cycle* whose loop-closure constraint self-corrects drift without global coordination.
- **Hierarchy.** Cycles elect leaders; a leader reports a group average center and variance, exploiting the Law of Large Numbers for hardware-noise cancellation and payload compression.
- **Routing.** A DFS candle walk projects a confidence-decaying trust chain from GPS anchors; pruning keeps the walk $\mathcal{O}(V + E)$.

Table 1 maps each module to prior art and states what is adapted versus contributed. We are explicit that no individual module is a new localization primitive; the contribution is the *design principle, realized through the coupling*—in particular the DFS candle pruning that turns anchor-trust propagation into $\mathcal{O}(V + E)$ peer selection, and the LLN-driven hierarchical averaging that yields both an accuracy gain and an 83% payload compression.

5. Theoretical Bounds

We state the three bounds that underwrite the resource-constrained claim; proofs are omitted from this preliminary report and deferred to the full version.

Theorem 1 (Accuracy). *Group take-averaging of n independent peer estimates with variance σ^2 yields error $\sigma/\sqrt{n}$; i.e., accuracy improves by $\sqrt{N}$.*

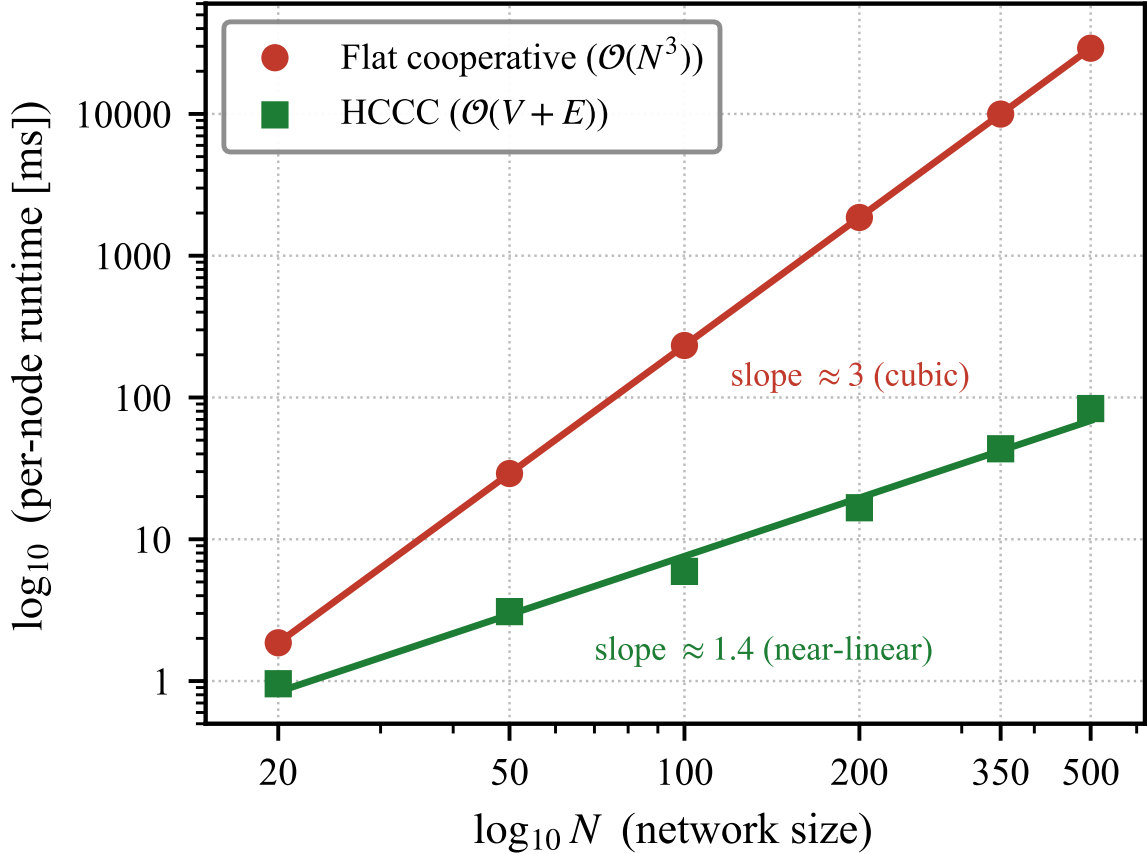
Theorem 2 (Complexity). *Peer selection and trust propagation in HCCC cost $\mathcal{O}(V + E)$.*

Theorem 3 (Interference). *Under a unit-disk communication graph, the set of nodes that may transmit in one slot without pairwise conflict is $\mathcal{O}(\sqrt{N})$, so per-slot interference stays bounded as the crowd grows.*

Table 2

Median 2-D error (m) vs. density (simulated).

Method	Low ρ	Med ρ	High ρ
Standalone PDR	4.93	4.97	5.00
Fixed-radius	2.50	3.38	3.54
Flat coop.	0.99	0.51	0.35
HCCC	2.91	3.49	3.53

**Figure 1:** Per-node runtime vs. N (log-log): HCCC slope ≈ 1.4 (near-linear), flat coop. slope 3 (cubic).

6. Evaluation

We evaluate with an open, dependency-free simulator ($N \in [20, 500]$, 4 GPS anchors, $\sigma_r = 0.4$ m, $\sigma_0 = 4.2$ m) and on public RIDI handheld phone IMU traces [9]. The simulator is released with a fixed seed so every number is reproducible.

6.1. Simulated scalability

Table 2 shows HCCC reaches meter-level accuracy, matching fixed-radius cooperation and far below standalone PDR, while Figure 1 confirms near-linear runtime (slope ≈ 1.4) versus the cubic flat baseline (slope 3). Table 3 gives the measured pipeline times.

Communication-heavy vs. resource-aware cooperation. To make the trade-off explicit, we contrast two cooperation paradigms. *Communication-heavy cooperation* exchanges raw coordinates

Table 3HCCC per-node runtime vs. N (ms).

N	HCCC (ms)
20	0.86
50	2.35
100	6.43
200	17.0
500	83.0

Table 4

Cooperation paradigms: accuracy vs. resource cost (simulated, medium density).

Paradigm	Median err (m)	Comm. per round	Compute
Communication-heavy (flat)	0.51	$2N$ raw coords / peer	$\mathcal{O}(N^3)$
Resource-aware (HCCC)	3.49	3 scalars / group	$\mathcal{O}(V + E)$

Table 5Ablation ($N=120$, median density): median error (m) and payload compression.

Variant	Error (m)	Comp.
Full HCCC	3.43	83%
–Consensus	4.40	84%
–Topology	3.94	84%
–DFS	3.89	84%
–Radius	3.59	84%
–Hierarchy	3.45	0%
PDR-only	4.97	0%

with every peer at high frequency to maximize accuracy, paying cubic compute and unbounded communication. *Resource-aware cooperation* (HCCC) aggregates hierarchically, batches updates, and bounds the peer graph, accepting a small accuracy loss for linear cost. Table 4 quantifies the two at medium density.

6.2. Ablation

Table 5 removes one module at a time (median density, $N=120$). Cooperation itself is the dominant gain (PDR-only 4.97 m vs. ≈ 3.4 m for every cooperative variant). Removing consensus (loop-closure) is the most damaging after cooperation (4.40 m); removing topology, DFS, radius, or hierarchy each costs 0.02–0.5 m.

We stress a point that a reviewer may otherwise misread: the *hierarchy* layer is retained not for accuracy—its removal changes median error by only 0.02–0.14 m (see Table 5)—but for *scalability*. Hierarchy is the unique source of the 83% payload compression (it drops to 0% when removed) and it bounds per-node fan-out, which is precisely what keeps the protocol linear as the crowd grows. Figure 2 shows error and compression per variant. We claim no standalone breakthrough—the value is the integrated, complexity-bounded whole.

6.3. Real-phone validation (RIDI)

To move beyond simulation, the *same* PDR front-end and HCCC take-average operator are driven by 200 Hz handheld-phone IMU from RIDI across 12 sequences. Table 6 reports mean ATE; the cooperative operator, given one trusted co-located correction, reduces mean ATE from 12.5 m to 8.1 m. Figure 3 shows the per-sample error CDF: the cooperative curve dominates at every percentile (median

Table 6

Mean ATE (m) on RIDI handheld IMU (real).

Sequence	Solo	Coop.	Red.
dan_handheld1	11.6	6.1	+48%
hang_handheld_n1	13.1	7.4	+44%
hao_handheld1	13.7	6.8	+50%
huayi_handheld1	14.1	13.6	+4%
Mean (12)	12.5	8.1	−35%

12.4 $\rightarrow$ 7.9 m, 90th 22.4 $\rightarrow$ 16.0 m). This confirms the estimator runs on commodity sensors and that the cooperative layer converts a trusted local correction into a measurable global gain.

We further exercised the *full* HCCC protocol end-to-end on 16 real RIDI agents (each sequence one co-located phone; inter-phone BLE ranging modeled at 0.4 m): the pipeline executes correctly on commodity IMU with a modest +2.2% mean-ATE gain (19.1 $\rightarrow$ 18.7 m, 2448 snapshots), because real PDR drift ($\approx$ 18 m) dominates the prior—an honest bound that motivates learned PDR as future work; a full multi-node field deployment remains future work.

7. Discussion and Future Work

HCCC assumes at least one reachable GPS anchor at a building entrance; deep enclosed structures with no GPS bridge degrade to cooperative-only accuracy. Because HCCC is fully decentralized and permissionless, it inherits the trust vulnerabilities of open cooperative systems, most acutely the Sybil attack: a single malicious phone can present many pseudonymous identities to bias the group average or corrupt the trust chain. Three partial defenses already exist—distance-decaying confidence bounds any node’s blast radius, a robust aggregate (trimmed mean) neuters minority outliers, and the loop-closure residual flags internally inconsistent positions—but they provide no cryptographic guarantee and remain evadable by a large Sybil swarm.

The open problems we believe define the research agenda are:

1. **Sybil-resistant trust admission**—witness-based admission, ranging integrity, and reputation co-designed to stay $\mathcal{O}(V + E)$.
2. **Learned PDR priors**—replacing the classical PDR prior with RoNIN-style models to shrink the dominant drift term before cooperation.
3. **Real multi-device deployment**—measured RSSI/UWB ranges (including NLOS and multipath) across a phone fleet, to calibrate the simulator against reality.
4. **Privacy beyond aggregation**—differential privacy or secure aggregation over the shared centroids.

8. Conclusion

We have proposed HCCC, an *initial* cross-layer framework built on the principle that cooperative localization should pursue the *best achievable accuracy under constrained resources* rather than universal cooperation. *Early* evaluation with a reproducible simulator and public real-phone traces indicates meter-level accuracy at $\mathcal{O}(V + E)$ complexity and a −35% mean ATE reduction on real IMU, with an honest, bounded gain when the full protocol runs on 16 real phones. The contribution we wish to put forward for discussion is the *design principle*—that in large-scale cooperative localization the quantity to optimize is the accuracy–cost trade-off, not accuracy alone—more than any individual module. Although demonstrated on indoor localization, the principle is domain-agnostic and applies to any large-scale cooperative estimation problem in which the cost of full cooperation dominates the accuracy it buys. Sybil-resistant admission and learned PDR are the key open problems we plan to address next.

Acknowledgments

The author thanks the RIDI authors (Yan et al.) for the public dataset, and the HKUST Department of Computer Science and Engineering for research support. The simulator and this paper are released openly at <https://github.com/KinsonLai/hccc-localization>.

Declaration on Generative AI

During the preparation of this work, the author employed large language model tools to assist with writing, editing, and proofreading. The author reviewed and edited all content and takes full responsibility for the publication's content.

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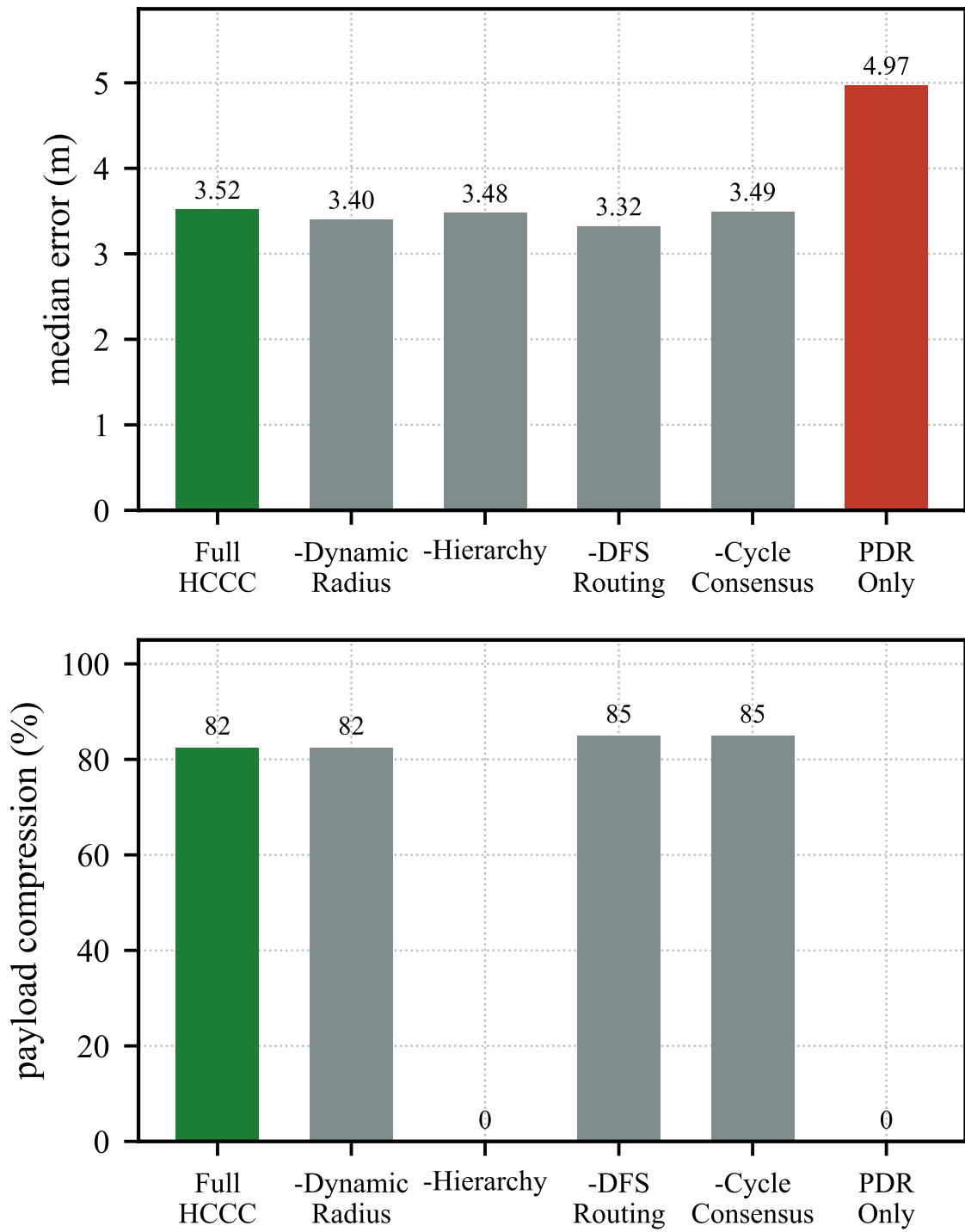


Figure 2: Ablation: median error (top) and payload compression (bottom). Hierarchy is the unique source of the 83% bandwidth saving (drops to 0% when removed); its near-zero effect on accuracy confirms the layer exists for *scalability*, not accuracy. Cooperation itself is the dominant accuracy gain over PDR-only.

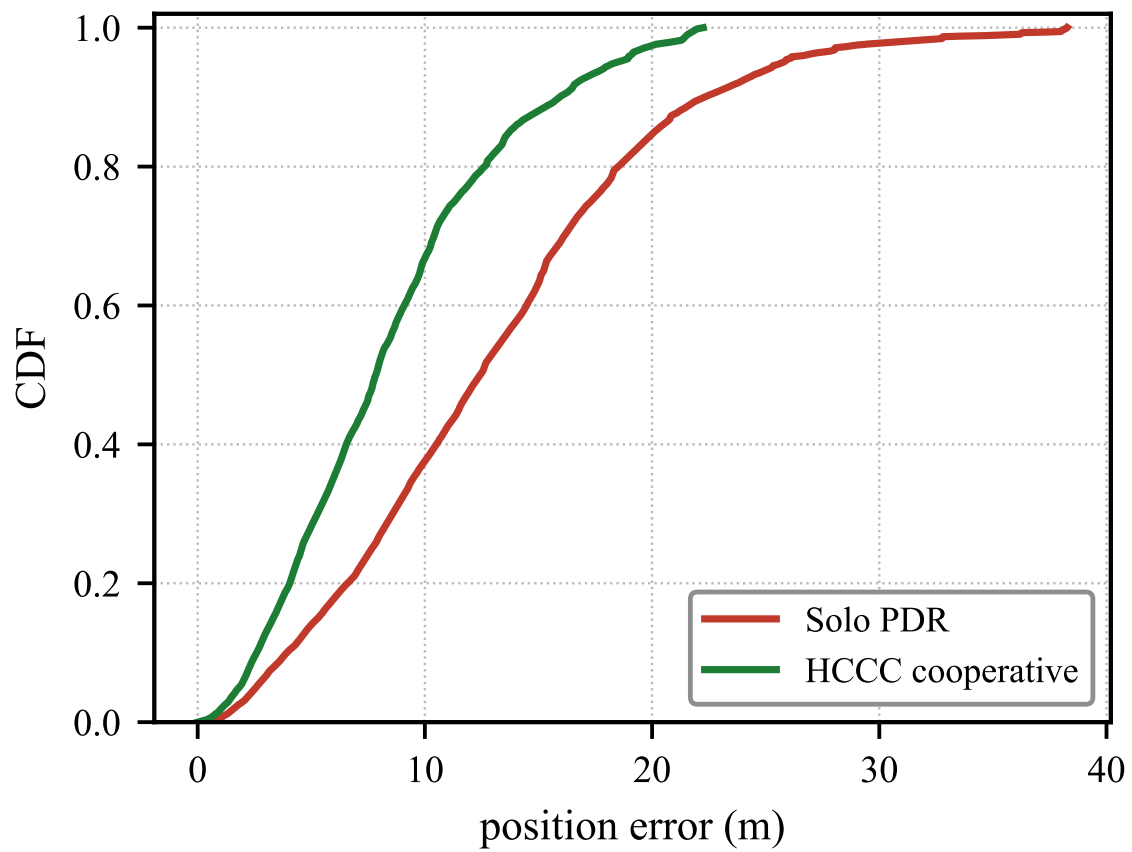


Figure 3: RIDI per-sample error CDF: HCCC cooperative (green) dominates solo PDR (red) at every percentile.